

11.1)

$$A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \\ -2 & 1 & 1 & 1 \end{pmatrix}$$

$$A \cdot A^T = A \cdot \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 7 & -2 & -2 \\ -2 & 7 & -2 \\ -2 & -2 & 7 \end{pmatrix}$$

VLASTNÍ ČÍSLA:

$$\lambda_1 = 9, \quad \lambda_2 = 9, \quad \lambda_3 = 3$$

SINGULÁRNÍ ČÍSLA:

$$\sigma_1 = 3, \quad \sigma_2 = 3, \quad \sigma_3 = \sqrt{3}$$

VLASTNÍ VECTORY PRO AA^T

PRO $\lambda = 3$

$$w_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

PRO $\lambda = 9$, VYBEREME ORTONORMÁLNÍ BAZI

$$w_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad w_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

SPÓČÍTAŤ ME PRÁVE SINGULÁRNÉ VEKTORY

$$N_i = \frac{A^T w_i}{\sigma_i}$$

$$N_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

$$N_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ 2 \\ -1 \\ -1 \end{pmatrix}$$

$$N_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

- PROTOŽE, A JE 3×4 , POTREBUJEME ŠESTÉ VEKTOR DO ORTONORMÁLNEJ BÁZE

$$Ax = 0$$

$$\begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 0 & -3 & 3 \\ 0 & -3 & 0 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & -3 & 0 & 4 \\ 0 & 0 & -3 & 3 \end{pmatrix}$$

$$x = (0, 1, 1, 1)$$

$$N_4 = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

$$V = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$$

SINGULAR: ROZKLADY

$$\underline{A = U D V^T}$$

$$\Rightarrow \underline{A^T = V D^T U^T}$$

$$A A^T = (U D V^T)(V D^T U^T) = U D (V^T V) D^T U^T = U D D^T U^T \\ \hookrightarrow V^T V = I$$

$$D D^T = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$A A^T = U \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix} U^T$$

$$A^T A = (V D^T U^T)(U D V^T) = V D^T D V^T \\ \hookrightarrow U^T U = I$$

$$D^T D = \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A^T A = V \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} V^T$$